

Welcome

Thank you for purchasing the Circuitly.in Light Following Robot DIY Kit.

This educational robotics kit helps students learn about:

- Arduino programming
- Sensor interfacing
- Motor control
- Robotics fundamentals
- Light sensing technology

Once assembled, the robot automatically detects and follows a light source such as a flashlight or mobile phone torch.

Safety Instructions

- Use the kit under adult supervision.
 - Do not short-circuit the battery terminals.
 - Keep electronic components away from water.
 - Disconnect power while assembling the robot.
 - Avoid looking directly into high-intensity light sources.
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Kit Contents:

Arduino UNO Compatible Board	1
L298N Motor Driver Module	1
LDR Sensor Module	2
Wheels	2
Bo Motor	2
Castor wheel	1
18650 cell	2
18650 cell holder	2
Connector wires	Assorted
Robot chasis	1
Mounting hardware	1kit
Charging module c type	1

What is a Light Following Robot?

A Light Following Robot is an autonomous robot that moves toward a light source.

The robot uses two LDR (Light Dependent Resistor) sensors placed at the front of the robot. These sensors continuously measure light intensity.

The Arduino compares both sensor readings and controls the motors accordingly:

- Left sensor detects more light → Robot turns left
 - Right sensor detects more light → Robot turns right
 - Both sensors detect equal light → Robot moves straight
 - No significant light detected → Robot stops or searches
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Understanding the LDR Sensor

LDR stands for Light Dependent Resistor.

Its resistance changes according to light intensity:

- More Light → Lower Resistance
- Less Light → Higher Resistance

This property allows the robot to determine the direction of a light source.

Assembly Instructions

Step 1: Assemble the Chassis

1. Fix both BO motors on the chassis.
2. Attach the wheels to the motors.
3. Install the castor wheel at the front.

Step 2: Mount Electronics

1. Fix the Arduino board on the chassis.

2. Mount the L298N motor driver.
3. Mount both LDR sensors on the front side.
4. Secure all modules using screws or cable ties.

Step 3: Connect Motors

Connect:

- Left Motor → Motor A Output
- Right Motor → Motor B Output

of the L298N module, Remove jumper of ENA and ENB available L298N motor driver

Step 4: Connect Sensors

Connect both LDR modules:

- VCC → 5V
- GND → GND
- Signal → Arduino Input Pins

Step 5: Connect Power

Connect the battery pack to the power input.

Verify all connections before powering ON.

Working Principle

The robot continuously monitors light intensity using two sensors.

Case 1: Light on Left Side

- Left sensor detects stronger light.
- Arduino slows/stops the left motor.
- Right motor rotates.
- Robot turns left.

Case 2: Light on Right Side

- Right sensor detects stronger light.
- Arduino slows/stops the right motor.
- Left motor rotates.
- Robot turns right.

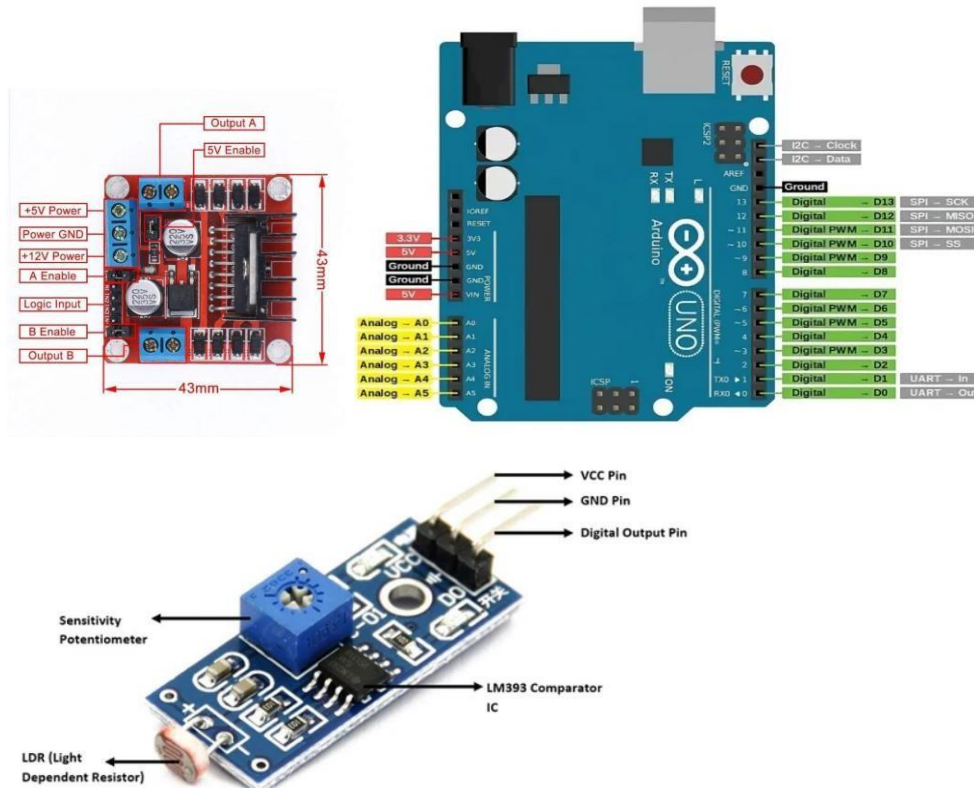
Case 3: Light in Front

- Both sensors detect similar light.
- Both motors rotate.
- Robot moves forward.

Pin connection with Arduino

L298N driver Pins	Arduino Pins
ENA	10
IN1	2
IN2	3
IN3	4
IN4	5
ENB	11
LDR Sensor	Arduino Pins
VCC	5V
GND	GND
Do(Right LDR)	7
DO(Left LDR)	8

Note : if motor rotate in opposite direction then try to change sensor pin



Uploading the Program

1. Install Arduino IDE.
2. Download link for Arduino <https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE#installation-instructions>
3. You can download code from <https://github.com/circuitlyin/SmartTrack-kit>
4. Watch the step by step guide video <https://www.youtube.com/@Circuitlyin>
5. Connect Arduino to PC using USB cable.
6. Select:
 - Board: Arduino UNO
 - Correct COM Port
7. Open the provided Circuitly.in code.
8. Click Upload.
9. Wait for "Upload Complete".

Testing the Robot

1. Switch ON the robot.
2. Place it on a flat surface.
3. Shine a flashlight in front.
4. Move the light source slowly.
5. Observe the robot following the light.

Troubleshooting

Robot Not Moving

- Check battery voltage.
- Verify motor connections.
- Confirm Arduino code upload.

Robot Moving in Wrong Direction

- Swap motor wires.
- Reposition LDR sensors.

Robot Not Detecting Light

- Check sensor wiring.
- Adjust LDR sensitivity potentiometer.
- Test with a brighter flashlight.

Motors Running Continuously

- Verify sensor values in Serial Monitor.
- Check Arduino program logic.

Learning Outcomes

After completing this project, students will understand:

- Arduino Basics
- Sensor Interfacing
- Motor Drivers
- Autonomous Robotics and Embedded Systems Fundamentals

Support

Circuitly.in

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Thank you for choosing Circuitly.in DIY Learning Kits.

Happy Building! Happy Learning!